Multi-task Learning for Commercial Brain Computer Interfaces

George Panagopoulos
gpanagopoulos@uh.edu , giorgospanagopoulos.github.io
Computational Physiology Lab ,University of Houston, Houston, Texas 77204

COMPUTATIONAL PHYSIOLOGY LAB
UNIVERSITY of HOUSTON
DEPARTMENT OF COMPUTER SCIENCE

Introduction

Problem
Improve subject generalization of passive economical Brain Computer Interfaces

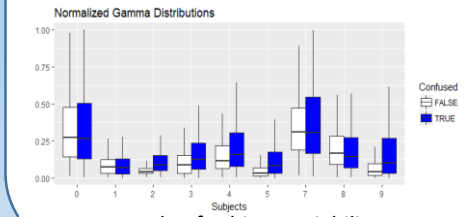
Motivation:
Exploit the trade off between noisy individual recordings and increased number of subjects

Advantages:
Utilize knowledge shared between subjects during training
Extract patterns present in all subjects instead of explicit individuals

Datasets

Berkeley Experiment [1]
- 30 subjects
- One 5-minute session each
- Two types of stimuli during session
  - Math, memorizing colors, think of items
  - Listen to music, watch video ads, relax
- Cognitive state changes in the same session
- Classes: Mental activity or relaxation

Carnegie Mellon Experiment [2]
- 9 subjects
- Ten 2-minute sessions with MOOC videos
- Self-classified levels of confusion for each session
- Cognitive state changes between sessions
- Classes: Confused or not



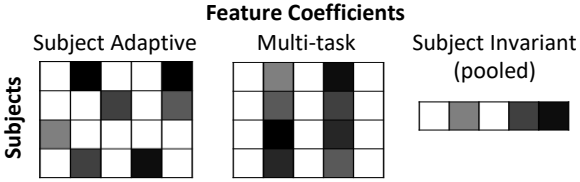
Example of subject variability

Methods

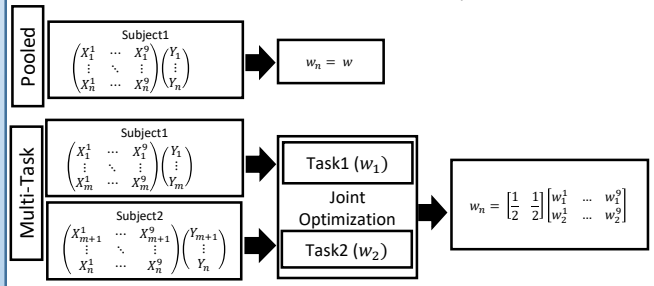
Discriminative MTL[1]
Logistic regression with group sparsity constraint on the K coefficients of all subjects T

$$min \frac{1}{N_t} \sum_{t=1}^{T} J(\mathrm{X_t}, \mathrm{W_t}, \mathrm{Y_t}) + \lambda \sum_{k=1}^{K} |W_k|_2$$

Rows of W are the feature coefficients w_t of each subject t
Each approach yields a different W:



The column average of multi-task W is the w_n of a new subject n



Generative MTL[2]
Bayesian estimation of the coefficients prior distribution, which is shared for all subjects

$$p(W; X, Y) \sim \prod_{t=1}^{T} p(y_t, X_t; w_t) p(w_t)$$

$$p(w_t) \sim N(\mu, \Sigma), \forall\, t \in T$$

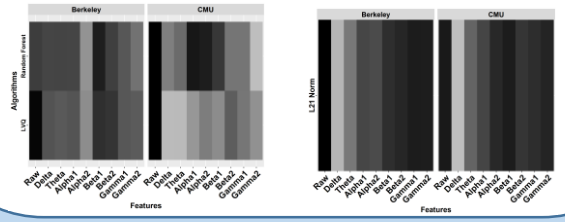
$$\min_{w_t, \mu, \Sigma} \frac{1}{\sigma^2} \sum_{t=1}^{T} ||X_t w_t - Y_t||^2 + \frac{1}{2} \sum_{t=1}^{T} (w_t - \mu)^T \Sigma^{-1} (w_t - \mu) + \frac{T}{2} logdet(\Sigma)$$

The sign of the regressed value determines the binary class.
For a new subject a coefficient vector w_n is sampled from $N(\mu, \Sigma)$

Results

Compare MTL methods with pooled approaches from the literature
- MTL methods perform better in the Berkeley dataset because it contains less subject specific information than CMU
- When the number of subject increases, MTL methods display substantial robustness in contrast with the rest of the approaches
- Consistent pattern extraction that agrees with the field's literature

	CMU (9 subjects)	Berkeley (10 subjects)	Berkeley (30 subjects)
Linear Discriminant Analysis	63.92	63.61	57.73
Shrinkage Linear Discriminant Analysis	63.92	63.61	57.73
Linear Support Vector Machine	64.16	64.15	56.21
Multi Layer Perceptron	58.43	55.63	45.90
Radial Basis Neural Network	35.84	35.85	48.71
Learning Vector Quantization	59.94	61.48	55.67
K Nearest Neighbors	55.21	54.48	53.03
Decision Tree	**64.28**	**64.27**	54.23
Random Forest	63.52	63.66	57.83
Extreme Boosting	63.14	61.43	55.27
Ensemble	62.69	59.91	54.34
Logistic Regression l_{21}	51.29	64.15	**64.16**
Bayesian Shared Prior	52.80	63.58	63.42



Conclusion

Multi-task algorithms are more robust than conventional pooled approaches to the subject generalization problem.
- Improved or steady accuracy with more subjects
- Consistent among datasets feature selection

References

[1]Argyriou, Andreas, Theodoros Evgeniou, and Massimiliano Pontil. "Convex multi-task feature learning." Machine Learning 73.3 (2008): 243-272.
[2] Alamgir, Morteza, Moritz Grosse-Wentrup, and Yasemin Altun. "Multitask learning for brain-computer interfaces." International Conference on Artificial Intelligence and Statistics. 2010.

Code

https://github.com/GiorgosPanagopoulos/Multi-task-Learning-for-Commercial-Brain-Computer-Interfaces

[1] https://www.kaggle.com/berkeley-biosense/synchronized-brainwave-dataset [2]https://www.kaggle.com/wanghaohan/eeg-brain-wave-for-confusion